IDC PERSPECTIVE

Leveraging GenAI in Healthcare: How Fujita Health University Hospital Transformed Clinical Workflows by Leveraging Amazon Bedrock

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EXECUTIVE SNAPSHOT

FIGURE 1

Executive Snapshot: Leveraging GenAI in Healthcare: How Fujita Health University Hospital Transformed Clinical Workflow by Leveraging Amazon Bedrock

This report showcases the GenAI case study from Fujita Health University (FHU) hospital in Japan, where GenAI is utilized to assist in the creation of discharge summaries. The case revealed that the use of GenAI significantly reduced the workload of clinicians. An approach grounded in management strategy and governance, was shown to be important for successful GenAI deployment.

Key Takeaways

- In 2023Q3, FHU launched a strategic initiative, with the initial focus on enhancing productivity by applying GenAI to the creation of discharge summaries.
- FHU partnered with Amazon Web Services (AWS) to conduct a proof of concept (PoC), leveraging their GenAI platform, Amazon Bedrock. By 2025 Q1, the project had transitioned to full-scale operation, and its use cases rapidly expanded.
- IDC survey highlighted that deploying AI and automation in healthcare operations is the topmost priority for the Asia/Pacific healthcare organizations in the next 12–24 months.
- The most important criteria for Asia/Pacific care providers when evaluating an AI technology partner are AI security capabilities and a cloud ecosystem partnership with AI expertise, as revealed by IDC survey data.

Recommended Actions

- Establish a GenAI governance system through a Clinical Oversight Committee comprising clinicians, data scientists, legal experts, and patient safety officers.
- Integrate GenAI into electronic health record (EHR) workflows for documentation automation, as IDC data shows that Japanese care providers prioritize EHR in their automation strategy.
- Establish robust data security by strengthening the hospital's data architecture, particularly EHRs.
- Define and prioritize GenAI use cases that align with clinical and operational goals. Involve line-of-business (LOB) leaders in collaborative discussions with the IT team.
- Work with trusted partners, selected through defined criteria in line with the project in-hand to ensure seamless and intuitive GenAI adoption. Also, evaluate their cloud ecosystem partnerships.

Source: IDC, 2025

June 2025, IDC #AP52884325

SITUATION OVERVIEW

Fujita Academy was established in 1964 under the founding philosophy of "Our Creativity for the People." This powerful philosophy drives everything, from pioneering clinical practice to cutting-edge research, and the cultivation of tomorrow's healthcare leaders. Fujita Health University (FHU), the educational institution of the Fujita Academy, operates four affiliated hospitals: Fujita Health University Hospital (FHUP), Bantane Hospital, Nanakuri Memorial Hospital, and Okazaki Medical Center. Among them, FHUP is one of the largest medical facilities in Japan, with 1,376 beds. Focusing on its social responsibility, Fujita Academy has established FUJITA VISION 2030 with the new slogan, "Becoming Fujita Academy as the First Responder," to tackle pressing social issues, including large-scale disasters and emerging infectious pandemics. Under this vision and its founding philosophy, FHU is collaborating with ecosystem partners, such as the healthcare industry, academia, and government, to realize the Smart Hospital concept and develop an advanced IT system, while promoting the healthcare digital transformation (DX).

This IDC study examines Fujita Academy and FHU's initiatives in healthcare DX, based on publicly available information and interviews conducted with the FHU. It specifically explores a case study on the use of Generative AI (GenAI) for creating discharge summaries.

Healthcare DX Trends Across Asia/Pacific

Healthcare provider organizations across the Asia/Pacific are undergoing a fundamental shift, not only in how care is delivered, but also in how healthcare operations are optimized and how the healthcare workforce experience is reimagined. Increasingly, the focus extends beyond traditional clinical services to encompass workflow efficiency, clinician productivity, and patient engagement, with the understanding that sustainable health systems, embedded with data and governance mechanisms, require innovation across all dimensions of care. The primary goal is to improve health outcomes by ensuring smarter resource allocation, enhancing diagnostic accuracy, and empowering clinicians and frontline staff to devote more meaningful time to patient interaction and decision-making.

Driven by rising patient volumes, aging populations, and workforce shortages, conventional care models are encountering limitations. Consequently, care providers in Asia/Pacific are actively exploring the transformative potential of AI, particularly, GenAI. These advanced technologies are emerging as strategic enablers, capable of automating documentation, supporting clinical decision-making, personalizing patient communication, and streamlining operational tasks. For seamless adoption and successful transitioning from pilot to production, all these use cases must be anchored in clinical data sets and supported by a robust governance mechanism.

FHU's Focus on Clinical Data

FHU holds an extensive repository of healthcare data, including 1.2 million electronic medical records (EMRs), health checkup data from 6,000 staff members, and real-world data (RWD) such as genomic information related to cancer and psychiatric conditions. To effectively leverage these assets, FHU is building a centralized data warehouse (DWH) to aggregate RWD from hospitals, clinics, and health checkup centers. The university aims to collaborate with companies and government to deliver valuable health insights to individuals by leveraging personal health record (PHR) applications and IoT devices. FHU established the Fujita Healthcare Platform (PHR platform) on AWS. After conducting a demonstration experiment with university staff, FHU is building a collaborative platform for the secondary use of RWD to benefit wider sections of society.

FHU's Focus on Governance to Promote Healthcare DX

FHU emphasizes that a strong management perspective is essential for achieving digital transformation in healthcare. To achieve this, it has actively strengthened its governance structure, and several dedicated departments have been established to drive this effort, namely,

- Digital Strategy Department, responsible for formulating overarching digital strategies:
- Smart Hospital Promotion Office, which focuses on implementing DX within hospital operations; and
- Health Data Architecture Center (HDAC), which promotes digital transformation in medical data, research, and the cultivation development of skilled professionals.

Leveraging GenAI to Address Critical Challenges

Supported by a robust clinical data platform, an integrated healthcare ecosystem, and strong data governance frameworks, FHU is well-positioned to leverage GenAI tools to address key healthcare challenges. In alignment with this vision, FHU partnered with AWS to implement a GenAI use case within its care delivery system, aiming to improve clinical workforce efficiency.

GenAI Case Study: How Fujita Health University Improved the Clinical Workflow by Leveraging GenAI

In Q3 2023, FHU launched a GenAI initiative, strategically focusing on low-risk, high-impact use cases to enhance productivity. In close collaboration with AWS, FHU conducted a thorough evaluation of governance frameworks, feasibility, implementation challenges, and overall comprehensiveness. Based on this assessment, the university identified the automated preparation of patient discharge summaries as a priority use case, with the purpose of significantly reducing clinicians' administrative burden. A proof of concept (PoC) was then

initiated to validate the approach and its potential impact in a real-world clinical setting.

Background: The Need for GenAI in Preparing Discharge Summaries

FHU recognized the value of using GenAI to reduce labor-intensive administrative tasks for clinicians. Among these, preparing discharge summaries, is a low-risk, high-impact use case. At FHU, discharge summary preparation demands an immense amount of time from hospital workforce and requires collaboration across multiple departments. Leveraging GenAI integrates various departments and is expected to drive work style transformation, streamline workflows, and reduce operational costs.

This GenAI use case, given the direct impact on improving clinician efficiency, gained strong support from the medical staff. To explore its real-world applicability, FHU's Digital Strategy Department launched a project leveraging AWS solutions to explore the application of GenAI in creating discharge summaries.

GenAI Solution: Amazon Bedrock

Building on its earlier collaboration with AWS to develop a PHR platform, FHU extended the partnership to transform discharge summary preparation. Discharge summaries are critical documents that record a patient's diagnosis, treatment, and recovery journey during hospitalization. Therefore, this involves data sets from various departments within a hospital across the patient care continuum. FHU launched this pilot project by leveraging Amazon Bedrock, a platform offering foundation models (FMs) via API. The project leveraged GenAI to create high-quality discharge summaries, significantly reducing the time clinicians spend on this task.

AWS not only provided technical expertise and infrastructure support, but also actively participated as a core member of the PoC team. The Digital Strategy Department led the project with support from FHU's IT and clinical departments, ensuring alignment with governance and compliance standards for effective deployment of GenAI solutions.

The GenAI model integrated both structured and unstructured clinical data, including non-labeled documentation. It did so without being directly embedded into existing systems, allowing for flexible data handling while avoiding disruption to core platforms. Patient data privacy and security were safeguarded through a policy-based system architecture that embedded data protection measures into every layer of the solution.

Challenges: Data Scarcity and Skill Gaps

During the PoC phase, FHU identified several critical challenges in deploying GenAI for automating the discharge summary process. Key challenges included limited

data availability, such as missing medical histories and incomplete treatment documentation, which affected the accuracy and completeness of the summaries generated. Additionally, the AI model occasionally misused clinical terminology or misrepresented critical details, such as treatment dates or diagnostic sequences, raising concerns about its reliability in clinical settings.

Another challenge was the skill gap in advanced AI technologies within the internal team, particularly in areas like prompt engineering and model fine-tuning. To address this, FHU partnered closely with AWS and leveraged external expertise to upskill its internal resources. Furthermore, in response to output inaccuracies, FHU took corrective steps including switching to a more appropriate large language model (LLM) and reinforcing prompt engineering strategies. Additionally, it is considering the use of retrieval-augmented generation (RAG) to improve contextual accuracy and citation of clinical data.

Outcome: Transition from PoC to Production with a 90% Reduction in Summary Preparation Time

The PoC project at FHU, powered by Amazon Bedrock, initially piloted in the Department of Respiratory Medicine. It created 50 discharge summaries, which were reviewed by physicians. Based on feedback from the physicians, FHU refined the data, fine-tuned prompts, selected the most suitable LLMs, and improved the user interface. As a result, the project brought about a transformative improvement in the efficiency of discharge summary creation. By leveraging GenAI for this task, FHU succeeded in reducing the time required to create a discharge summary per patient from approximately 10 minutes to just 1 minute—a reduction of up to 90%. This remarkable reduction freed up valuable clinician time, allowing greater focus on patient communication and personalized care, ultimately improving both care quality and workload management.

Clinicians reported increased satisfaction due to the workstyle transformation, and there were measurable improvements in the consistency and quality of documentation. While the PoC revealed various challenges, it also identified solutions, demonstrating that the initiative is both practical and effective. In other words, discharge summary preparation has been recognized as a valuable use case for the application of GenAI.

Following the success of the PoC, FHU launched the discharge summary generation system powered by Amazon Bedrock into full-scale operation. In Q1 2025, the solution was scaled across more than 30 medical departments. This rapid expansion was made possible by leveraging FH-Hub, which enables the integration of dispersed medical information across the hospital while maintaining reliability, along with strong collaboration with partners such as AWS.

Key Features of Amazon Bedrock Suitable for Healthcare

In hospital environments like FHU in Japan, flexibility, robust privacy protection, strong data security, and a scalable architecture that aligns seamlessly with FHU's digital strategy are all essential. Amazon Bedrock offers a powerful foundation for deploying GenAI in healthcare by enabling access to a variety of foundation models through a single API, eliminating the need to manage underlying infrastructure. It also includes built-in capabilities to support strong privacy and data security measures, meeting the stringent requirements of medical institutions. Furthermore, AWS offers comprehensive guidance to ensure compliance with Japan's Two Guidelines from Three Ministries, supporting safe and responsible AI implementation in healthcare.

This PoC utilized the AWS Prototyping Program, a free service that helps customers develop prototypes of the systems they intend to build on AWS. By developing a prototype, customers can gain deeper insights into how AWS services work. This service covers everything from architecture design to UX design tailored to FHU's requirements, thereby accelerating POC development.

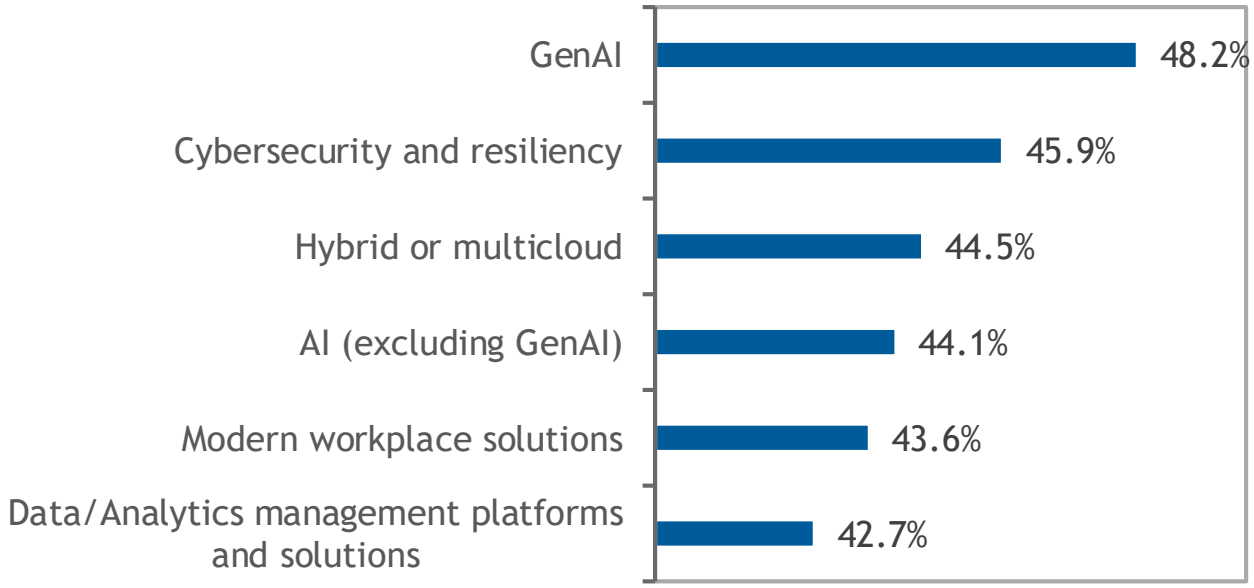
Amazon Bedrock not only offers fine-tuning capabilities for foundation models but also supports RAG, enabling integration with an organization's proprietary data. RAG allows models to reference resources such as guidelines for creating discharge summaries without the need for fine-tuning, significantly enhancing the accuracy their creation. Furthermore, the Amazon Bedrock Agents feature, along with Amazon Q, a virtual assistant, can further automate the management of system operations and workflows. Fujita Health University is currently considering the adoption of these capabilities.

IDC's Point of View

According to IDC's *Asia/Pacific Public Sector Survey, 2024*, almost half (48%) of healthcare provider organizations are investing in or planning to invest in GenAI in the next 12–18 months, as shown in Figure 2.

FIGURE 2

Technologies That Asia/Pacific Healthcare Organization Are Using and/or Planning to Invest in over the Next 12–18 Months



Note: n = 220

Source: IDC's *Asia/Pacific Public Sector Survey, 2024*

This new wave of technology is set to make AI accessible for a larger patient population, directly or indirectly. As a result, this GenAI transition is poised to transform the healthcare ecosystem as healthcare providers focus on maximizing workflow efficiency and healthcare outcomes.

The FHU case demonstrates how a project transitions from PoC to full-scale deployment in a short period, accompanied by a rapid expansion of its scope of application. In fact, at leading medical institutions, the use of GenAI has already become an integral part of their comprehensive digital strategies.

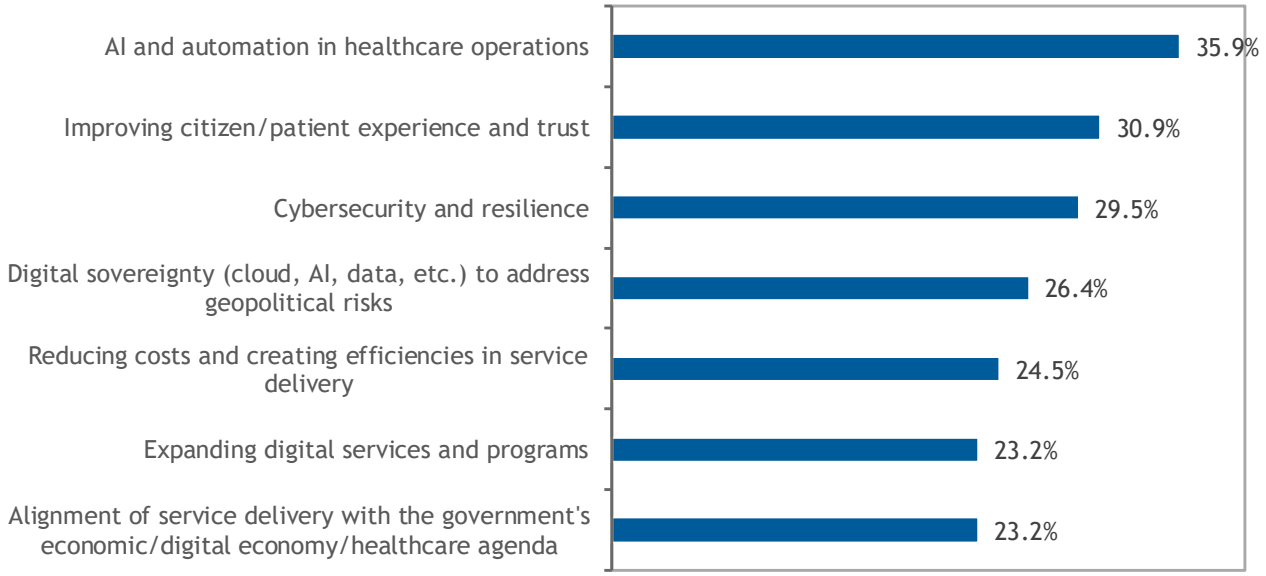
Operational Priorities

IDC's *Asia/Pacific Public Sector Survey, 2024* highlighted that the topmost priorities for the regional healthcare providers over the next 12–24 months are the adoption of AI and automation, improving patient experience, and enhancing cybersecurity, as shown in Figure 3. As many regions such as Singapore, South Korea, Taiwan, and Australia are witnessing surge in clinical data, AI and automation are seen as accelerators for scaling services in resource-constrained settings. Driven by competition in private healthcare markets and rising consumer expectations, there is growing pressure to improve the patient experience, which demands more accessible and affordable care. At the same time, increasing digitized and AI-based

use cases are expanding the attack surface, making cybersecurity a higher priority than ever.

FIGURE 3

Top Operational Priorities for the Care Providers in Asia/Pacific over the Next 12–24 Months



Note: n = 220

Source: IDC's *Asia/Pacific Public Sector Survey, 2024*

In contrast, the same survey revealed that for the majority of Japanese healthcare provider organizations, creating efficiency in service delivery and improving employee experience stand as the top operational priorities in the next 12–24 months. This closely aligns with the intended goal of the FHU case study detailed in this report.

As Japanese hospitals face mounting pressures from an aging population and a shrinking healthcare workforce, care providers are making employee experience and operational efficiency key strategic priorities. This shift reflects not only a response to staffing shortages, but also a recognition that improving frontline working conditions is essential for sustainable care delivery. GenAI is emerging as a powerful enabler in this context, streamlining administrative tasks, automating documentation, and simplifying communication workflows. Tools powered by GenAI can draft clinical summaries, generate patient discharge instructions, and support multilingual translation, dramatically reducing the cognitive and clerical burden on healthcare workers.

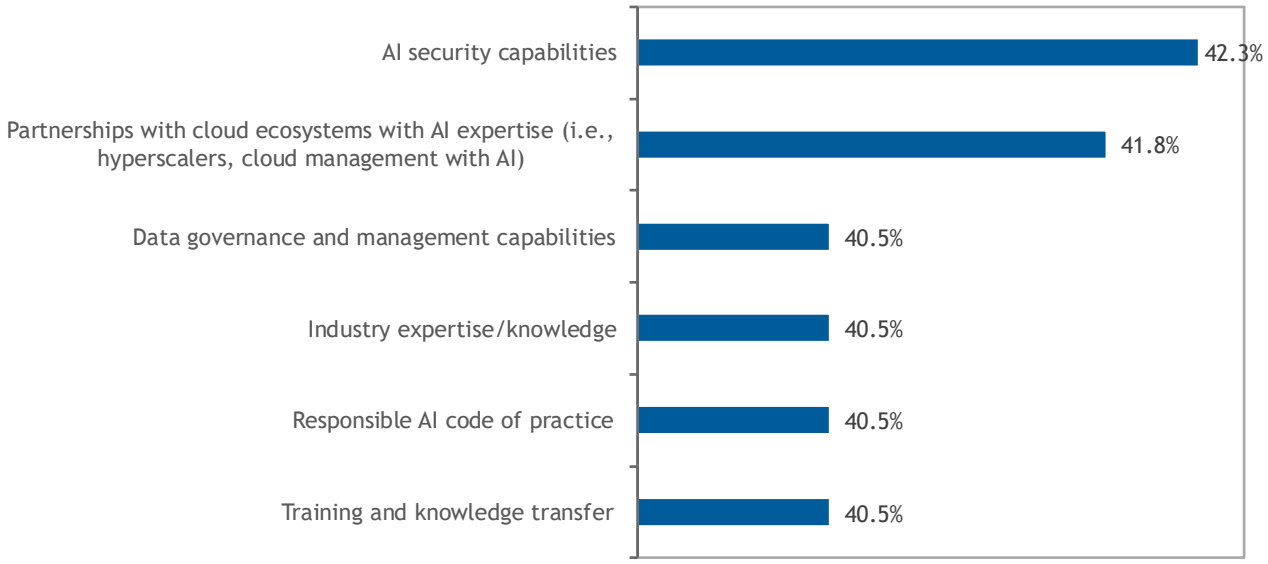
Given its immense potential in tackling such care needs, GenAI is being integrated into care management systems across Japanese hospitals to support decision-making, personalize treatment pathways, and assist with real-time triage, enabling clinical staff to deliver more effective care with fewer resources. By focusing on both operational efficiency and employee satisfaction, Japanese hospitals are positioning GenAI not just as a productivity tool, but as a catalyst for transforming the workplace experience in healthcare. This shift contributes to building a more resilient care system.

Preferred Vendor Attributes

According to IDC's *Asia/Pacific Public Sector Survey, 2024*, AI security capabilities and partnerships with cloud ecosystems with AI expertise topped the vendor selection criteria for healthcare providers, as shown in Figure 4. Many regions are rapidly digitizing healthcare, often leapfrogging to cloud-based infrastructure. As a result, partnerships with major cloud players who offer AI-ready services are crucial.

FIGURE 4

Criteria for Evaluating or Selecting an AI Technology Provider



Note: n = 220

Source: IDC's *Asia/Pacific Public Sector Survey, 2024*

However, the majority of healthcare providers in Japan are becoming increasingly selective in choosing AI technology partners, placing a strong emphasis on technology and knowledge transfer, AI security, and responsible AI practices. Given Japan's aging population and its push for digital transformation in healthcare, providers are not just looking for plug-and-play AI tools, but are seeking long-term

collaborators who can build local capacity and empower clinical staff with the skills and understanding needed to manage and evolve AI solutions independently. Hence, they prioritize "Training and knowledge transfer" as the topmost criteria for evaluating an AI partner, as revealed by IDC data.

Moreover, due to strict data privacy regulations and the sensitive nature of health data, Japanese care providers prioritize vendors with proven AI security capabilities, including data encryption, model explainability, and robust risk mitigation strategies. The growing awareness of algorithmic bias and ethical implications of AI has also made it essential for vendors to adhere to responsible AI codes of practice, such as transparency, fairness, and human oversight.

Key Learning Points

This in-depth analysis of FHU's GenAI case study highlights key insights that can serve as guiding principles for healthcare providers planning to implement GenAI solutions within their organizations. The top three key takeaways are,

Adhere to a Robust Governance Mechanism. According to IDC *Asia/Pacific Public Sector 2024* data, one in four healthcare providers in Japan view the lack of responsible AI governance as a significant risk in the implementation and management of AI systems. In this case study, FHU clearly emphasized it was actively strengthening its governance structure through dedicated departments such as the Digital Strategy Department and the Smart Hospital Promotion Office to set a strong foundation.

Align Use Cases and Engage Both the Digital/IT Team and Lines of Business (LOBs). FHU strategically aligned its GenAI use case with one of its most persistent operational challenges—discharge summary preparation, a time-intensive task for clinicians. The project was effectively led by FHU's Digital Strategy Department, ensuring active collaboration between Digital/IT and clinical teams. This cross-functional involvement was critical in maintaining alignment with governance, compliance, and operational goals.

Close the Loop to Measure Impact and Drive Continuous Improvement. FHU emphasized the principle that "what gets measured, gets improved." To validate and refine its GenAI implementation, FHU conducted clinician workload assessments and satisfaction surveys, which confirmed the solution's positive impact. By closing the feedback loop, FHU leadership identified increased clinician satisfaction driven by workstyle transformation, along with measurable improvements in the consistency and quality of discharge documentation, enabled by a standardized, GenAI-powered approach.

ADVICE FOR THE TECHNOLOGY BUYER

Successful deployment of GenAI solutions within a care provider organization depends upon multiple factors, including having a competent tech partner, strong data governance, integration with clinical records, and robust data security.

- **Work with Trusted Technology Partners.** Select the right technology partner to ensure that GenAI adoption is seamless and user-friendly. Healthcare organizations should prioritize working with competent, experienced AI vendors that have a proven track record in the healthcare industry. As revealed by IDC data, key partner evaluation criteria should include the capability to support technology transfer, ensure AI security, follow responsible AI practices, and engage in cloud ecosystem partnerships to demonstrate AI expertise.
- **Establish a GenAI Governance and Clinical Oversight Committee.** Establish a cross-functional internal committee that includes clinicians, data scientists, legal advisors, and patient safety officers, to guide GenAI strategies, evaluate use cases, ensure model transparency, and maintain clinical alignment. This ensures that GenAI tools are deployed responsibly, meet Japan's healthcare regulations, and are clinically validated before integration into workflows such as radiology reporting or discharge summary preparation.
- **Integrate GenAI into Electronic Health Record Workflows for Documentation Automation.** IDC data shows that the top priority for the Japanese care providers is electronic health record (EHR) as part of their automation strategy. Therefore, GenAI can be integrated into existing EHR systems to assist clinicians with high-volume documentation tasks including clinical notes, discharge summaries, and referral letters.
- **Establish a Strong Foundation for Data Security and Infrastructure.** Since GenAI tools are heavily data-driven, hospitals should begin by strengthening their data architecture, particularly the EHR systems. Ensure that EHR systems are modernized and interoperable with GenAI platforms to enhance both functionality and UX. Invest in scalable public or hybrid cloud infrastructure equipped with AI-optimized compute capabilities and built-in advanced cybersecurity measures such as end-to-end encryption to protect sensitive patient data throughout AI workflows.
- **Define and Align GenAI Use Cases Across the Organization.** With the data and infrastructure in place, clearly identify and prioritize GenAI use cases that align with clinical and operational goals. Involve LOB leaders, such as clinical heads and operational managers, in collaborative discussions with the CIO and technology partners to ensure alignment and adoption. Focus areas may include improving clinician productivity through AI-driven documentation tools, enhancing patient engagement via intelligent chatbots, or streamlining workflows through automated decision support systems.

- **Ensure Compliance with National Data Protection Laws.** Asia/Pacific countries are actively advancing AI strategies and road maps for the next five to ten years, alongside healthcare data protection and data sharing guidelines. Aligning with these strategies and regulations will foster synergy in GenAI implementation within care provider settings.

LEARN MORE

Related Research

- *Smart Aging: The Rise of AI-Driven Elderly Care Ecosystem* (IDC #AP52884425, June 2025)
- *Industry Market Trends: Asia/Pacific Healthcare, 2025* (IDC #US52218625, May 2025)
- *Industry Market Forecast: Asia/Pacific Healthcare, 2025* (IDC #US52218925, May 2025)
- *Fortifying the Healthcare Sector in Asia/Pacific: The Essential Role of AI in Cybersecurity* (IDC #AP52113325, March 2025)
- *IDC FutureScape: Worldwide Healthcare Industry 2025 Predictions — Asia/Pacific (Excluding Japan) Implications* (IDC #AP51588124, December 2024)
- *GenAI in Healthcare and Life Sciences: Current Trends and Future Potential in Asia/Pacific* (IDC #AP51587824, June 2024)

Synopsis

This IDC Perspective showcases the GenAI case study from Fujita Health University in Japan. The initiative focuses on reducing physicians' administrative burden by automating the generation of hospital discharge summaries. By implementing this solution, FHU achieved a 90% reduction in the time required to complete each summary. Following a successful pilot, the project has now progressed into a scalable implementation phase, showcasing strong potential for broader adoption.

"Healthcare providers across Asia/Pacific have identified GenAI as a top technology investment priority over the next two years. To ensure a successful transition from pilot to production, and to realize the full value of GenAI adoption, several key prerequisites must be addressed. These include selecting a capable and trusted technology partner, establishing a strong data governance framework, aligning GenAI use cases with strategic organizational goals, and upskilling internal teams to effectively adopt and manage AI-driven workflows," says Manoj Vallikkat, senior research manager, IDC Asia/Pacific Health Insights.

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